Chapter 7. The Sieve

You Shall Not Overcount

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7.1 Enumerating the elements of intersecting sets

Motivation. Adding the sizes of overlapping sets counts common elements several times. Alternately subtracting and adding back the sizes of intersections corrects every overcount exactly; the reason is that $\sum_j (-1)^j \binom{s}{j} = 0$.

Example 7.1

14 students play soccer, 17 basketball, and 4 both. Then $14 + 17 - 4 = 27$ play at least one of the two.

Example 7.2

Add 18 hockey players; 4 play soccer and basketball, 3 soccer and hockey, 5 basketball and hockey, and 1 all three. Then $14 + 17 + 18 - 4 - 3 - 5 + 1 = 38$ students play at least one game. The student playing all three is counted three times, subtracted three times, and must be added back once.

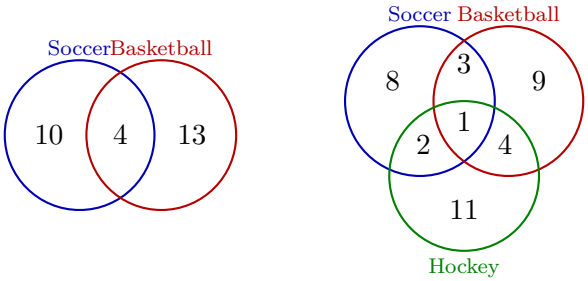


Figure. *Examples 7.1 and 7.2: the counts in the regions of the Venn diagrams sum to* 27 *and* 38.

Theorem 7.3 (Sieve formula, inclusion–exclusion)

For finite sets $A_1, \dots, A_n$,

$$|A_1 \cup \cdots \cup A_n| = \sum_{j=1}^{n} (-1)^{j-1} \sum_{\{i_1,\dots,i_j\}} |A_{i_1} \cap \cdots \cap A_{i_j}|, \tag{7.1}$$

where $\{i_1, \dots, i_j\}$ runs over the j-element subsets of $[n]$.

Proof. An element outside the union is counted by no term. Let x be in the union, $S := \{i \mid x \in A_i\}$ and $s := |S| \geq 1$. An intersection $A_{i_1} \cap \cdots \cap A_{i_j}$ contains x iff $\{i_1, \dots, i_j\} \subseteq S$, so x is counted

$$s - \binom{s}{2} + \binom{s}{3} - \cdots + (-1)^{s-1} \binom{s}{s} = 1 - (1-1)^s = 1 \tag{7.2}$$

times, by the binomial theorem. ∎

7.2 Applications of the sieve formula

Motivation. Counting objects with *none* of several bad properties is the typical use of the sieve: subtract the objects with at least one bad property, which inclusion–exclusion counts through intersections that are easy to count. Derangements and surjections are the classic cases; the second gives the promised formula for $S(n, k)$.

Example 7.4 (Derangements)

n guests pick up their hats at random. In how many ways does nobody get their own hat, i.e. how many permutations of $[n]$ have no fixed point (**derangements**)?
Solution. Let A_i be the set of permutations fixing i. An intersection of j of the A_i fixes j given points and permutes the rest freely, so it has $(n-j)!$ elements, and there are $\binom{n}{j}$ such intersections. By (7.1),

$$|A_1 \cup \cdots \cup A_n| = \sum_{j=1}^{n} (-1)^{j-1} \binom{n}{j} (n-j)! = \sum_{j=1}^{n} (-1)^{j-1} \frac{n!}{j!},$$

so the number of derangements is

$$D(n) = n! - |A_1 \cup \cdots \cup A_n| = \sum_{i=0}^{n} (-1)^i \frac{n!}{i!}. \tag{7.3}$$

Hence $\frac{D(n)}{n!} = \sum_{i=0}^{n} \frac{(-1)^i}{i!} \to e^{-1}$: roughly 37% of all permutations are derangements. [Supplement] In fact $D(n)$ is the integer nearest to $n!/e$ for $n \geq 1$: the alternating tail $\left|\frac{n!}{e} - D(n)\right| = n! \left|\sum_{i>n} \frac{(-1)^i}{i!}\right|$ is at most its first term $\frac{n!}{(n+1)!} = \frac{1}{n+1} \leq \frac{1}{2}$, with strict inequality since the terms decrease strictly.

Theorem 7.5

For positive integers n, k:

$$S(n,k) = \frac{1}{k!} \sum_{i=0}^{k} (-1)^i \binom{k}{i} (k-i)^n.$$

Proof. By Corollary 5.9, $k!\,S(n,k)$ is the number of surjections $[n] \to [k]$. Of the k^n functions, let A_i be those missing i in their image; an intersection of j such sets consists of functions into a $(k-j)$-element set, so it has $(k-j)^n$ elements. By the sieve, the number of non-surjective functions is $\sum_{j=1}^{k} (-1)^{j-1} \binom{k}{j} (k-j)^n$. Subtract this from k^n. ∎

Theorem 7.6 (Möbius inversion on subsets)

Let f, g be real functions on the subsets of $[n]$ with $g(S) = \sum_{T \subseteq S} f(T)$. Then

$$f(S) = \sum_{T \subseteq S} g(T) (-1)^{|S-T|}. \tag{7.4}$$

Proof. Expanding each $g(T)$, the right side of (7.4) is $\sum_{U \subseteq S} f(U) \sum_{U \subseteq T \subseteq S} (-1)^{|S-T|}$. A set T with $U \subseteq T \subseteq S$ is determined by $S \setminus T \subseteq S \setminus U$, and there are $\binom{|S-U|}{i}$ choices with $|S-T| = i$. So the inner sum is $\sum_i (-1)^i \binom{|S-U|}{i} = (1-1)^{|S-U|}$, which is 0 unless $U = S$, when it is 1. Hence the right side is $f(S)$. ∎

(This is the prototype of Möbius inversion on partially ordered sets.)

7.3 Selected exercises

Motivation. The book exercises below apply the sieve to number theory and to set partitions, derive the dual form of Möbius inversion, and give a bijective proof of the derangement recurrence; the first one is a warm-up in elementary counting.

Exercise 7.2 (Three teams)

A class of at least two students has three sports teams, and any two students are both members of at least one team. Prove that some team contains at least $\frac{2}{3}$ of the students.

Solution. Let $n \geq 2$ be the number of students and T_1, T_2, T_3 the teams.

Step 1 (a student in only one team). Suppose some student a belongs to exactly one team, say T_1. For every other student b, some team contains both a and b; it can only be T_1, so $b \in T_1$. Hence T_1 contains all students.

Step 2 (no student in only one team). Every student a is in some team, since a shares a team with another student. If no student is in exactly one team, every student is in at least two teams, so $|T_1| + |T_2| + |T_3| \geq 2n$, and the largest team has at least $\frac{2n}{3}$ students. ■

Exercise 7.8 (Euler's function)

Let $\varphi(n)$ be the number of integers in $[n]$ relatively prime to n. Prove that if $p_1, \ldots, p_k$ are the distinct prime divisors of n, then

$$\varphi(n) = n\Big(1 - \frac{1}{p_1}\Big) \cdots \Big(1 - \frac{1}{p_k}\Big).$$

Solution.

Step 1 (the sets). Let $A_i = \{x \in [n] : p_i \mid x\}$. An integer x is relatively prime to n iff no p_i divides x, so $\varphi(n) = n - |A_1 \cup \cdots \cup A_k|$.

Step 2 (intersections). For $\emptyset \neq S \subseteq [k]$ let $d_S = \prod_{i \in S} p_i$. Since the p_i are distinct primes, x lies in all A_i, $i \in S$, iff $d_S \mid x$. As $d_S \mid n$, exactly n/d_S elements of $[n]$ are multiples of d_S.

Step 3 (sieve). By Theorem 7.3, with $d_\emptyset = 1$,

$$\varphi(n) = n - \sum_{\emptyset \neq S \subseteq [k]} (-1)^{|S|-1} \frac{n}{d_S} = \sum_{S \subseteq [k]} (-1)^{|S|} \frac{n}{d_S} = n \prod_{i=1}^{k} \Big(1 - \frac{1}{p_i}\Big),$$

the last step by expanding the product: choosing $-\frac{1}{p_i}$ from the factors $i \in S$ and 1 from the others gives the term of S. ■

Exercise 7.12 (The dual Möbius inversion)

Let h, r be real functions on the subsets of $[n]$ with $r(S) = \sum_{T \supseteq S} h(T)$ for all S. Prove that

$$h(S) = \sum_{T \supseteq S} r(T)(-1)^{|T-S|}.$$

Solution. Write $\bar{X} = [n] \setminus X$; complementation is a bijection of the subsets of $[n]$ that reverses inclusion.

Step 1 (transport). Define $f(A) = h(\bar{A})$ and $g(A) = r(\bar{A})$. Then, substituting $T = \bar{U}$,

$$g(A) = r(\bar{A}) = \sum_{T \supseteq \bar{A}} h(T) = \sum_{U \subseteq A} h(\bar{U}) = \sum_{U \subseteq A} f(U).$$

Step 2 (apply Theorem 7.6). Hence $f(A) = \sum_{U \subseteq A} g(U)(-1)^{|A-U|}$ for all A.

Step 3 (transport back). Take $A = \bar{S}$ and substitute $U = \bar{T}$; then $U \subseteq A$ iff $T \supseteq S$, and $A - U = \bar{S} \cap T = T - S$. So

$$h(S) = f(\bar{S}) = \sum_{T \supseteq S} g(\bar{T})(-1)^{|T-S|} = \sum_{T \supseteq S} r(T)(-1)^{|T-S|}.$$

■

Exercise 7.23 (The derangement recurrence)

Give a combinatorial proof of $D(n+1) = n(D(n) + D(n-1))$ for $n \geq 1$, without using (7.3). Here $D(0) = 1$ and $D(1) = 0$.

Solution. Let p be a derangement of $[n+1]$ and $k = p(n+1)$; then $k \in [n]$, which gives n choices. Fix k; we show that exactly $D(n-1) + D(n)$ derangements have $p(n+1) = k$.

Step 1 ($p(k) = n+1$). Then $(k\ n{+}1)$ is a 2-cycle of p, and the other cycles form a derangement of the $(n-1)$-element set $[n] \setminus \{k\}$. Conversely every such derangement, together with the 2-cycle, gives one p. This gives $D(n-1)$ derangements.

Step 2 ($p(k) \neq n+1$). Then the cycle of $n+1$ has length at least 3. Delete $n+1$ from its cycle: with $m = p^{-1}(n+1)$, let $p'(m) = k$ and $p'(x) = p(x)$ for the other $x \in [n]$. Then p' is a permutation of $[n]$, and it has no fixed point: $p'(m) = k \neq m$ because $m \neq k$ (as $p(k) \neq n+1$), and the other values are those of p.

Step 3 (the inverse of Step 2). Given a derangement q of $[n]$, insert $n+1$ before k in its cycle: with $m = q^{-1}(k)$ put $p(m) = n+1$, $p(n+1) = k$, and $p(x) = q(x)$ otherwise. This is a derangement of $[n+1]$ with $p(n+1) = k$ and $p(k) = q(k) \neq n+1$, and the two constructions are inverse to each other. This gives $D(n)$ derangements.

Summing over the n values of k, $D(n+1) = n(D(n)+D(n-1))$. (For $n = 1$: $D(2) = 1 \cdot (0+1) = 1$.)
■

Exercise 7.37 (No singleton blocks, by the sieve)

Let $F(n)$ be the number of partitions of $[n]$ without singleton blocks. Express $F(n)$ through the Bell numbers.

Solution. We show $F(n) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} B(n-j)$, with $B(0) = 1$.

Step 1 (the sets). Let A_i be the set of partitions of $[n]$ in which $\{i\}$ is a block. Then $F(n) = B(n) - |A_1 \cup \cdots \cup A_n|$.

Step 2 (intersections). For a j-element set $J \subseteq [n]$, the partitions in $\bigcap_{i \in J} A_i$ consist of the singletons $\{i\}$, $i \in J$, and an arbitrary partition of the $n-j$ remaining elements. So the intersection has $B(n-j)$ elements, and there are $\binom{n}{j}$ such J.

Step 3 (sieve). By Theorem 7.3,

$$F(n) = B(n) - \sum_{j=1}^{n} (-1)^{j-1} \binom{n}{j} B(n-j) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} B(n-j).$$

For example $F(3) = 5 - 3 \cdot 2 + 3 \cdot 1 - 1 = 1$, the single partition $\{1,2,3\}$. Compare Exercise 5.34, which relates F and B bijectively. ■